EXPERIMENT 7

Read a captured video and produce slow-motion and fast-motion versions.

Aim

Read a captured video and produce slow-motion and fast-motion versions.

Algorithm

• Import OpenCV and open the sample video.

• Read the video's frame rate and dimensions.

• Write every frame to the slow-motion video at a lower frame rate.

• Skip alternate frames for the fast-motion video.

• Save both processed videos and representative frames.

Code

```
import cv2

cap = cv2.VideoCapture("images/sample_video.avi")
if not cap.isOpened():
    raise FileNotFoundError("images/sample_video.avi not found")
fps = cap.get(cv2.CAP_PROP_FPS) or 15
w = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
h = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fourcc = cv2.VideoWriter_fourcc(*"MJPG")
slow = cv2.VideoWriter("output_images/experiment07_slow.avi", fourcc, max(1, fps / 2), (w, h))
fast = cv2.VideoWriter("output_images/experiment07_fast.avi", fourcc, fps * 2, (w, h))
frames = []
while True:
    ok, frame = cap.read()
    if not ok:
        break
    frames.append(frame)
    slow.write(frame)
cap.release()
for i, frame in enumerate(frames):
    if i % 2 == 0:
        fast.write(frame)
slow.release()
fast.release()
cv2.imwrite("output_images/experiment07_input_frame.png", frames[0])
cv2.imwrite("output_images/experiment07_output_frame.png", frames[-1])
print("Saved slow-motion and fast-motion videos")
```

Input Image:

Output Image:

Input Image

The input is a sample video supplied in the project folder.

Output Image

Frame 1